Department of Leather Engineering

Khulna University of Engineering & Technology

B. Sc. Engineering 4th Year 1st Term Examination-2022

LE 4109

Leather Products Engineering III

Time: 3.0 Hours | Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a)	"Culture and communication sets out to investigate how culture communicates"?- Explain briefly.	14
1(b)	What is the notion of Anthropologist Malinowski in case of protection?	07
1(c)	Mention popular clothing for men and women in two distinguished ancient eras Egypt and Gothic.	08
1(d)	Mention the main features of a sailor collar.	06
2(a)	Mention the significance of Blouson jacket. Define and specify with related figures for the following terminologies: **(i)** Revere, **(ii)** Body rise, **(iii)** Cape length and **(iv)** Fall.	15
2(b)	During jacket block construction why front neck curve more concave than back neck curve?	07
2(c)	Define stand collar and compare between standard convertible collar and shawl collar.	08
2(d)	Explain with sketch-why BPP of a gents bodice block is higher than FPP?	05
3(a)	Define front opening system. Briefly explain the following front opening systems: **(i)** Button with button hole and **(ii)** Snap button.	10
3(b)	Illustrate the techniques of lower the armhole and extending the shoulder during sleeve enlargement.	15
3(c)	Calculate leather consumption for one-piece Blouson jacket in sq.ft.	10
4(a)	Develop a formula for calculating length and waist radii of women cloth.	07
4(b)	How can we give a fullness effect to the skirt block fabrication?	12
4(c)	Define a Slit and mention the functions of a Pleat.	05
4(d)	If the waist girth and skirt length of a women are 63cm and 53.7cm respectively then determine waist and skirt length radii for one, two and four panel circular skirt.	11

SECTION–B

5(a)	Define perspective drawing. Design a waist coat and sketch the complete set of patterns construction of your designed waistcoat.	16
5(b)	Mention the methods of pattern making. Note the principal measurements for developing a waistcoat.	10

5(c)	Illustrate the functions of major anatomy points on waistcoat block.	05
6(a)	Why Velcro is the salient choice as fastener than that of metal pin buckle? Explain.	07
6(b)	Describe the complete back preparation technique of a trouser with related figures.	15
6(c)	Briefly state the principle of standard fittings for a garment.	08
6(d)	Define square shoulder problem of a waistcoat. How do you solve it?	05
7(a)	How do accessories differ from fashion accessories and trimmings?	10
7(b)	Construct a list of multiple fasteners and fittings for assembling a haversack.	06
7(c)	Define molded article. State the materials used for the stiffening of leather goods.	12
7(d)	If a consumer has a waist of 35 inches, calculate his approximate chest and half across back.	07
8(a)	Remark on brand and brand equity. State the major parameters to measure brand strength.	13
8(b)	What is large abdomen fitting problem of a trouser? How can you minimize it?	08
8(c)	Write down the salient features of the followings: **(i)** Zero point perspective, **(ii)** Pelerine, **(iii)** Liberty cap and iv) Fashion draping.	14

Department of Leather Engineering

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B. Sc. Engineering 4^{th} Year 1^{st} Term Examination-2022

LE 4111

Footwear Engineering III

Time: 3.0 Hours　　　　Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a)	What is meant by Swedish strip steel knife? Discuss the classification of this knife.	10
1(b)	Briefly explain the production process of Swedish strip steel knife.	08
1(c)	Illustrate the graphical method for material allowance calculation with its benefits and limitations.	10
1(d)	Define 1^{st}, 2^{nd} and 3^{rd} wastes based on Russ and Small method.	07
2(a)	What is meant by laser cutting technology? "Laser cutting method in footwear industry offers huge advantages and some drawbacks too"- explain it.	09
2(b)	Mention the needle, thread and stitch specifications while joining the derby upper with lining part in case of leather upper and lining material.	06
2(c)	Discuss the points to be checked during attaching topline tape in the shoemaking process.	08
2(d)	Write short notes on: (**i**) U-bound topline (**ii**) Bagged topline and (**iii**) Strobel stitching.	12
3(a)	Write down the top ten golden rules for closing department.	10
3(b)	What are the checkpoints if thread breaks during stitching? What are the possible strategies to mitigate those problems?	08
3(c)	Why is it necessary to use desiccants during packing shoes? Discuss briefly the commonly used desiccants in footwear industry.	07
3(d)	Draw a shoe and indicate the common quality problems observed during quality inspection.	10
4(a)	What is article number? State the significance of article number of footwear.	05
4(b)	Define the shoe specification sheet. Why is it necessary to use the material map in the spec sheet?	05
4(c)	Write short notes on: (**i**) Depreciation cost (**ii**) Royalty cost and (**iii**) Semi variable cost.	09
4(d)	"Shoe materials are very susceptible to moisture"–explain it relating to warehouse storage and shipment.	08
4(e)	Discuss briefly airborne hazards that are produced in production areas in footwear industries.	08

SECTION–B

5(a)	Write down the complete sequence of operations in lasting department.	15
5(b)	Why cellulose board insole is widely used as insole material for footwear production?	10
5(c)	Write short notes on the following terms: **(i)** Insole recessing **(ii)** Insole bevelling **(iii)** Green strength of an adhesive and **(iv)** Tack life of an adhesive.	10
6(a)	Why temporary attachment by adhesive is needed during shoe lasting process.	10
6(b)	Suppose you are in lasting department; during lasting the following problems are occurred. Write down the cause and possible remedies of the following problems: **(i)** Lasted margin lifted **(ii)** Damaged insole **(iii)** Poor upper tension all around and **(iv)** Excess material enter into pincers.	10
6(c)	Define vulcanized shoe construction. Discuss the sequential operation of the direct vulcanizing shoe construction process.	10
6(d)	Write down the advantages of injection moulding process of PVC.	05
7(a)	Write short notes on the following manufacturing operations of Veldtschoen shoe construction: **(i)** Seat flanging **(ii)** Rough runners and **(iii)** Edge pairing.	12
7(b)	Differentiate among Californian Slip lasted construction with other available lasting construction technique.	08
7(c)	Describe the following terms for Californian Slip lasted construction: **(i)** Force lasting and **(ii)** Pull up platform cover.	10
7(d)	Write short note on string lasting process with figure.	05
8(a)	Define shoe finishing. Mention the objectives of shoe finishing	08
8(b)	Briefly explain the method of wrinkle chasing with their disadvantages.	12
8(c)	Discuss briefly about different types of top dressing agent.	10
8(d)	Write down the different types of cleaning agents with respect to their associated materials needed to be clean.	05

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B. Sc. Engineering 4th Year 1st Term Examination-2022

LE 4029

Tannery Waste Management

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a) Write down the sources and environmental effects of the following waste parameters generated in tannery effluent: (i) Chlorides (ii) Sulfides and (iii) BOD_5. 09

1(b) Categorize the tannery wastewater. State the key considerations for the treatment of tannery effluent. 08

1(c) Write about the unit operation involves in the removal of large solids from tannery wastewater with neat sketches. 08

1(d) Which beamhouse operation has highest pollution load? Justify your opinion. 05

1(e) Give the discharge standard of different waste parameters of tannery effluent into inland water based on ECR, 2023. 05

2(a) When the concentration of solid is not very high in a wastewater treatment plant then which coagulation principles is applied? Write about the mechanism of that principle. 08

2(b) Write about the zeta potential and its effect on the coagulation process using Stern's model of electrical double layer. 09

2(c) A tannery is planning to install a coagulation-flocculation treatment plant using alum coagulation. The average flow rate of the wastewater is 1.5 m^3/s. A laboratory Jar test showed the optimum result for 10s detention time in rapid mix tank with the velocity gradient of $1100S^{-1}$. The largest available mixing device in the market is JTQ 1000 having power of 7.46 KW with the rotational speed of 110 rpm and it works at 85% efficiency in water. Turbine type, 6 curved blades impeller with the impeller constant of 4.80 is only available in the market. Now design the coagulation tank and size the mixing equipment. [Dynamic viscocity of water, $\mu=1.053\times10^{-3}$ Pa.S; density of water, $\rho=10^3$ kg/m^3]. 12

2(d) State the role of alkalinity of water in coagulation-flocculation process. 06

3(a) Schematically show the instrumentation of activated sludge process and state the advantages and limitations of the system. 08

3(b) Write down the significance of the following operational parameters in controlling the activated sludge process: (i) F/M ratio (ii) MLSS and (iii) SVI. 09

3(c) Illustrate the liquid recirculation process in trickling filter with neat sketch. 08

3(d) A tannery wastewater treatment plant uses a single stage rock-media trickling filter for secondary treatment. The wastewater flow rate is 1800 m^3/d with a BOD_5 concentration of 350 mg/L. Primary clarification removes 25% of the BOD_5. The filter is 14m in 10 diameter and 1.8m in depth. Direct recirculation pump operates at 2.88 m^3/min to the filter. Wastewater temperature is 20°C. Calculate the hydraulic loading rate, organic loading rate, effluent BOD_5 concentration and overall plant efficiency.

4(a) Distinguish among different types of sedimentation process. 08

4(b) A tannery may treat about 12000 m^3/d of wastewater. Flocculation particles are produced by coagulation and a column analysis indicates that an overflow rate of 25 m/d will produce satisfactory removal at a depth of 3 m. Determine the size of the require rectangular settling tank. Also, find the retention time and weir loading rate. 10

4(c) Design a jar test experiment using alum coagulant for optimizing pH for tannery wastewater. 09

4(d) Write on the performance of different membrane process of wastewater treatment with schematic diagram. 08

SECTION–B

5(a) What is composite? State the role of reinforcement for composite material. 07

5(b) Explain the environmental impact of the chemical constituents of tannery solid wastes. 14

5(c) How can the Cr shaving/splitting be utilized as gelatin and poultry feed? 08

5(d) State the need of solid waste management in the tanning industries of Bangladesh. 06

6(a) State the conventional model for safely disposal of tannery sludge as landfill with proper diagram. 10

6(b) Discuss the major criteria of matrix for the preparation of composite material. 08

6(c) Define fiber reinforced composite. Why use the leather fiber for the preparation of composite material? 08

6(d) Describe the utilizations of the following materials **(i)** buffing dust **(ii)** Keratin and **(iii)** Finishing trimmings. 09

7(a) What is waste to energy (W to E)? Explain the technology of biogas production from tannery solid waste. 10

7(b) Define sustainable development. State the components of sustainable development in tanning industry. 08

7(c) State the toxicity of heavy metal components present in the tannery solid waste. 07

7(d) Write short notes on **(i)** Bio-chemical conversion and **(ii)** Composting. 10

8(a) What is cleaner technology? Why cleaner production is essential for sustainable development of tannery industry? 07

8(b) How can minimize the water consumption in tanning industry? 08

8(c) Write down necessary procedures for recycle or reuse of **(i)** Salt form curing and **(ii)** Used lime liquor. 08

8(d) Briefly discuss the recovery or reuse of chromium from chrome tanning wastewater by chemical precipitation process with proper diagram. 12

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Khulna University of Engineering & Technology

B. Sc. Engineering 4th Year 1st term Examination-2022

LE 4121

Environmental Science & Engineering

Time: 3.0 Hours — Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a) Define niche. Describe the niche interrelationship and material flow for both aquatic and terrestrial ecosystem. 10

1(b) What is eutrophication and cultural eutrophication? During hot weather, water body of a pond is covered with green algal bloom. What are the reasons behind excess algal growth in water? Also write down the solutions to remove algal bloom from that water body. 12

1(c) Calculate hydroxide, carbonate, and bicarbonate alkalinity of a water sample with a total alkalinity of 350 mg/L as $CaCO_3$ and pH 7.2. Which type of alkalinity dominates total alkalinity? 13

2(a) Analyse the mechanism of acid rain and global warming. 12

2(b) Write down the sources, standard values and significance of the following water quality parameters: **(i)** Nitrate **(ii)** Alkalinity **(iii)** Iron and **(iv)** Chlorides. 12

2(c) A water has the following analysis: data are given on table. Find the total hardness, Carbonate hardness and non-carbonate hardness. 11

Ion	Unit(mg/L)	Ion	Unit (mg/L)
Na^+	15	Cl^-	35
K^+	25	HCO_3^-	6.6
Ca^{2+}	07	CO_3^{2-}	10
Mg^{2+}	10	SO_4^{2-}	06
Sr^{2+}	02	NO_3^-	15

3(a) Define tannery solid waste. Write down the impacts of tannery solid waste on environment. 10

3(b) Describe the process of biofuel production from tannery solid waste. 10

3(c) 6 mL of bacterial culture is added to a 60 mL sterile diluent. From this suspension two serial of 1000 dilutions are made and 0.2 mL is added onto agar media from the last dilution. After incubation, 140 colonies are counted on the plate. Calculate CFU/mL of the original sample. 10

3 (d) Write down the significance of hardness in respect of tannery. 05

4(a) For BOD determination, 8 mL of waste water in mixed with 292 mL DI water with 9.1 mg/L DO. After 5 days incubation at 20°C, DO come 3.5 mg/L. calculate the BOD of the wastewater. Assume that initial DO of wastewater is zero. 15

4(b) Write down the mechanism of determining BOD value and what is the standard BOD value of tannery wastewater? 12

4(c) Write down the dangerous effect of lead and arsenic poisoning on human body. 08

SECTION–B

5(a) Discuss the importance of Environmental Evaluation system. 10

5(b) A EI point of 100 amongst some parameters have been given below: 25

Surface water (59)	Socio-economic (33)	Atmosphere (23)
BOD (25)	Water supply (15)	Air quality (23)
COD (08)	Navigation (18)	
TDS (11)		
TSS (15)		

Find out whether a project is feasible or not. (Assume, EQ range of value function curve=1-10)

6(a) Suppose, a tannery industry is supposed to be built nearby a river, away from the localized people. An ETP has been planned to treat the generated wastewater before releasing to the environment. Some experts have been brought to opinionate if it is feasible to build this there or not based on following parameters: 28

Parameters	EIA values	Parameters	EIA values
DO	15	Employment	28
BOD	20	Noise Pollution	12
COD	16	Fisheries	18
TDS	09	Water quality	19
TSS	11	-	-

Draw the value function curves of each parameter, find at EQ before and after project (by assuming) and show if the project is feasible or not.

6(b) Describe the COSHH guidelines for hazard control. 07

7(a) What are Terms of Reference (ToR)? "Mitigation should not reactionary response to environmental impacts" Explain. 08

7(b) Write down the significance of expert judgment in EIA. Assume with sketches the value function curve for dissolve oxygen for the Vairab River for the following conditions: **(i)** Present DO in the river is 6.8 mg/L **(ii)** After disposing water from nearby industries, the DO would be 3.08 mg/L. 13

7(c) Mention the discharge standards set by the ECR 2023 for the waste from tannery industry. 07

7(d) Differentiate IEE and EIA. 07

8(a) Define conservation of environment. Write down the procedures for issuing environmental clearance for "red" category project. 11

8(b) Define ecotoxicology and behavioral scheme of toxicants with organism. 09

8(c) Explain the role of baseline studies in EIA. 07

8(d) An alternative matrix for a tannery industry is given with impact value and importance in the following table. Determine major activities and define area which needs more attention by Leopold and from Lahani and Thanh. (Assume priority values from 1-10). 08

Proposed action / Impacts on	Plant location	Plant const.	Houling operation	Dust emission	Wastewater discharge	ETP	Employ-ment	Envir. Monit.
Agriculture	8/9					9/8	11/19	
Surface drainage		6/5		3/4				4/5
Air quality			6/9			4/5		
Water quality			11/15		12/8		7/8	
Human health		12/18			22/25			9/8
Fisheries			9/18				11/15	
Socio-economic condition		1/5			5/16			7/9
Navigation			2/9			16/19		

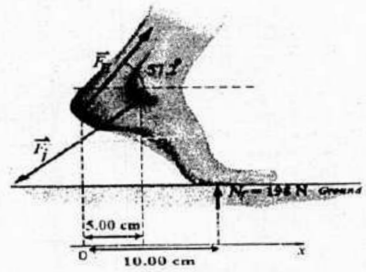
Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 4th Year 1st Term Examination - 2022
LE 4123
Computer Integrated Manufacturing Systems

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a)	What is meant by CAE? State the necessity of CIM towards factory modernization.	08
1(b)	Clarify the terms- M_{LT} and 4IR. Cite the elements of CIM wheel. What are the challenges for the implementation of CIMS in Bangladeshi footwear industry?	10
1(c)	"CIM involves the integration of quality progression with minimum cost and time" – explain with examples.	05
1(d)	A custom-made leather goods industry requires more than 3.0 weeks fabricating 330 units of leather bags. Recently, the industry has been acclimatization to automation that reduces the average set-up time from 3.0 hr. to 2.5 hr., and the average operation time per machine is 5.0 min to 4.0 min. but the sequences of eight operations remain constant in the plant. The average non-operation time due to handling delays, inspections and so on, is 6.0 hr. Now compute how many days it will take to produce the same units of bags and how do you explain the status of M_{LT}, assume the plant operates on a two-shift per day and each shift is 7.0 hr.	12
2(a)	Define pattern engineering. Show the step-by-step procedure of pattern engineering through shoemaster power software.	10
2(b)	Define NC, CNC and DNC machines and distinguish among them.	08
2(c)	What are the building blocks of a smart manufacturing system? Briefly state their functions.	07
2(d)	Figure out a 2D digitizer. Write the sequential command for mirror and add allowance of a footwear pattern by power software.	10
3(a)	Define ergonomic footwear with examples.	05
3(b)	What is meant by smart nesting? Mention some software which are widely used for leather garments.	08
3(c)	Mention the components of ground reaction force on 2D phase. Prove that $GRF_{Tol} = \frac{1}{2} mg [4\sin(4\pi t) - \sin(8\pi t)]$; where the symbols have their usual meanings.	10
3(d)	The father of the modern sprinter shoe was Adlof Dassler who began making shoe in 1920. To give support to the runner Dassler added three side strips technology to the shoe which assured foot safety to some extent. The data for a sprinter both as barefoot and puma shod running was recorded. The force during puma shod was 225N. The barefoot condition data for the sprinter just before toe off at the start of the race was shown in the following figure. First, draw the lever diagram and also find: (i) Force $\lvert\overrightarrow{F_H}\rvert$ exerted on the heel point by calcaneal tendon; **(ii)** Estimate variation of the force between bare and shod condition **(iii)** Magnitude of the force $\lvert\overrightarrow{F_J}\rvert$ exerted on the foot at the ankle joint.	12

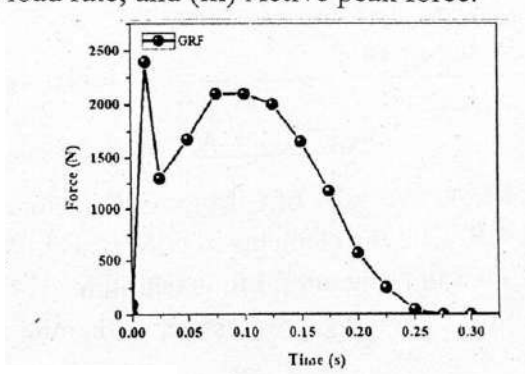


4(a)	Cite a few conventional software used in footwear designing.	06

4(b) Mention the use of modification tools in power software. 06

4(c) Describe the wire-frame model and solid model of footwear with related figures. 11

4(d) List out the necessary properties of insole that keep in mind during insole designing. The vertical GRF of a subject running with a sports shoe is given in figure. Estimate (**i**) Impact peak force (**ii**) Vertical load rate, and (**iii**) Active peak force. 12



SECTION – B

5(a) Define industrial automation. What are the salient reasons behind automation? 08

5(b) Build a relationship between a category of production and diversification of products. 07

5(c) Draw a schematic diagram that represents the relation between automation and CIM. 12

5(d) How does industrial automation affect the demand equation? 08

6(a) What is FMS? How various types of manufacturing systems can affect manufacturing system design? Briefly explain. 15

6(b) Write short notes on: (**i**) Job shop (**ii**) Project shop (**iii**) Flow shop and (**iv**) Continuous process. 20

7(a) Define fixed-position, process and product layout and mention their characteristics including material handling equipment. 10

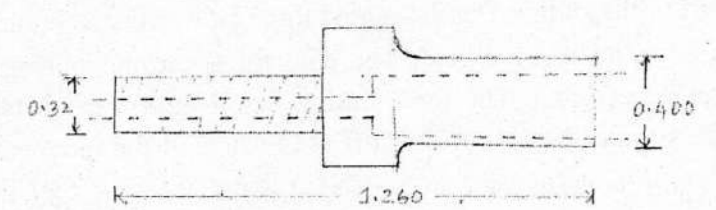
7(b) What is conveyor? Write down the following types of conveyors: (**i**) Roller conveyor (**ii**) Flat conveyor (**iii**) In-floor towline conveyor and (**iv**) Overhead trolley conveyor 16

7(c) If the available time is AT and delivery cycle time is T_c. So, what is the rate of delivery $R_{(dv)}$ per vehicle per hour? 04

7(d) Why are AGV systems used in a contemporary manufacturing plant? 05

8(a) Define part family. Briefly state the criteria that qualify a manufacturing system is flexible 10

8(b) Develop the form code (first 5 digits) in the Opitz system for the part illustrated in the figure. 10



8(c) Apply the rank order clustering technique to the part-machine incidence matrix in the following table to identify logical part families and machine groups. Parts are identified by numerically, and machines are identified by letter. 15

	Part Numbers						
Machine ID		1	2	3	4	5	6
	A			1		1	
	B		1	1			
	C	1			1		
	D		1	1		1	
	E	1			1		1